

Postgresql18

November 17, 2025

0. Informations Vos requêtes SQL jusqu'à 10000 fois plus rapides, durablement.

Sécurisons les accès à PostgreSQL

1. Réseau Podman dédié

```
podman network inspect pgnet
```

```
"gateway": "10.10.0.1",  
"subnet": "10.10.0.0/24"
```

2. Lancer les 3 containers PostgreSQL pg18_a (IP 10.10.0.10 – port exposé 5433)

```
sudo podman run -dt --name pg18_a \  
  --network pgnet --ip 10.10.0.10 \  
  -e POSTGRES_PASSWORD=admin \  
  -e POSTGRES_USER=postgres \  
  -p 5433:5432 \  
  postgres:18
```

pg18_b (IP 10.10.0.11 – port exposé 5434)

```
sudo podman run -dt --name pg18_b \  
  --network pgnet --ip 10.10.0.11 \  
  -e POSTGRES_PASSWORD=admin \  
  -e POSTGRES_USER=postgres \  
  -p 5434:5432 \  
  postgres:18
```

pg18_c (IP 10.10.0.12 – port exposé 5435)

```
sudo podman run -dt --name pg18_c \  
  --network pgnet --ip 10.10.0.12 \  
  -e POSTGRES_PASSWORD=admin \  
  -e POSTGRES_USER=postgres \  
  -p 5435:5432 \  
  postgres:18
```

3. Autoriser les connexions entre containers

À l'intérieur du réseau Podman, tous écoutent sur 5432.

À l'extérieur (hôte), on a :

```

- pg18_a → 5433
- pg18_b → 5434
- pg18_c → 5435

postgresql.conf :
vi +/listen_addresses /var/lib/postgresql/18/docker/postgresql.conf

listen_addresses = '*'
port = 5432

pg_hba.conf
vi /var/lib/postgresql/18/docker/pg_hba.conf

host      all      all      10.10.0.0/24      trust

```

4. Reload PostgreSQL dans chaque container

```

podman exec -u postgres pg18_a pg_ctl reload -D var/lib/postgresql/18/docker
podman exec -u postgres pg18_b pg_ctl reload -D var/lib/postgresql/18/docker
podman exec -u postgres pg18_c pg_ctl reload -D var/lib/postgresql/18/docker

```

5. Connexions entre les containers

Depuis pg18_a vers pg18_b :

```
psql -h 10.10.0.11 -U postgres -p 5432
```

Depuis pg18_a vers pg18_c :

```
psql -h 10.10.0.12 -U postgres -p 5432
```

Depuis l'hôte :

```
* pg18_b sur 5434 :
psql -h localhost -U postgres -p 5434
```

```
* pg18_c sur 5435 :
psql -h localhost -U postgres -p 5435
```

6. Sauvegarde simple

Depuis l'hôte :

```

mkdir /home/durandf/sauvegardes
cd /home/durandf/sauvegardes
podman exec pg18_a pg_dump -U postgres dvdrental > dvdrental.sql

```

7. Restauration sur pg18_b (port: 5434)

Depuis l'hôte :

```

psql -h localhost -U postgres -p 5434
postgres=# create database dvdrental;
ou

```

```
podman exec -it pg18_b psql -U postgres -c "CREATE DATABASE dvdrental;"
cd /home/durandf/sauvegarded
podman exec -i pg18_b psql -U postgres -d dvdrental < dvdrental.sql
```

8. Vérification de la restauration sur pg18_b

```
[5]: !sudo podman exec -it pg18_a psql -U postgres -d dvdrental -c "\dt"
```

```

                List of tables
 Schema |      Name      | Type | Owner
-----+-----+-----+-----
 public | actor           | table | postgres
 public | address          | table | postgres
 public | category         | table | postgres
 public | city             | table | postgres
 public | country          | table | postgres
 public | customer         | table | postgres
 public | film             | table | postgres
 public | film_actor       | table | postgres
 public | film_category    | table | postgres
 public | inventory        | table | postgres
 public | language         | table | postgres
 public | payment          | table | postgres
 public | rental           | table | postgres
 public | staff            | table | postgres
 public | store            | table | postgres
(15 rows)
```

```
[3]: !sudo podman exec -it pg18_b psql -U postgres -d dvdrental -c "\dt"
```

```

                List of tables
 Schema |      Name      | Type | Owner
-----+-----+-----+-----
 public | actor           | table | postgres
 public | address          | table | postgres
 public | category         | table | postgres
 public | city             | table | postgres
 public | country          | table | postgres
 public | customer         | table | postgres
 public | film             | table | postgres
 public | film_actor       | table | postgres
 public | film_category    | table | postgres
 public | inventory        | table | postgres
 public | language         | table | postgres
 public | payment          | table | postgres
 public | rental           | table | postgres
 public | staff            | table | postgres
```

```
public | store          | table | postgres
(15 rows)
```

9. Restauration sur pg18_c (port: 5435)

Depuis l'hôte :

```
psql -h localhost -U postgres -p 5435
```

```
postgres=# create database dvdrental;
```

ou

```
podman exec -it pg18_c psql -U postgres -c "CREATE DATABASE dvdrental;"
```

```
cd /home/durandf/sauvegardes
```

```
podman exec -i pg18_c psql -U postgres -d dvdrental < dvdrental.sql
```

10. Vérification de la restauration sur pg18_c

```
[4]: !sudo podman exec -it pg18_c psql -U postgres -d dvdrental -c "\dt"
```

List of tables			
Schema	Name	Type	Owner
public	actor	table	postgres
public	address	table	postgres
public	category	table	postgres
public	city	table	postgres
public	country	table	postgres
public	customer	table	postgres
public	film	table	postgres
public	film_actor	table	postgres
public	film_category	table	postgres
public	inventory	table	postgres
public	language	table	postgres
public	payment	table	postgres
public	rental	table	postgres
public	staff	table	postgres
public	store	table	postgres

(15 rows)

```
[ ]: %pip install ipython-sql
```

```
[6]: %load_ext sql
```

```
[ ]: %sql postgresql+psycpg2://postgres:admin@localhost:5433/dvdrental
```

```
[7]: %sql postgresql://postgres:admin@localhost:5433/dvdrental
```

```
[8]: %sql
SELECT oid, datname FROM pg_database WHERE datname='dvdrental';
```

```
* postgresql://postgres:***@localhost:5433/dvdrental
1 rows affected.
```

```
[8]: [(16775, 'dvdrental')]
```

11. Affichage de la base physique dvdrental sur pg18_a

```
[12]: !sudo podman exec pg18_a ls -C var/lib/postgresql/18/tbs_dvdrental/
↳ PG_18_202506291/16775
```

112	16874	17046	2616_vm	2702	3439	4149
113	16874_fsm	17047	2617	2703	3440	4150
1247	16885	17048	2617_fsm	2704	3455	4151
1247_fsm	16886	17049	2617_vm	2753	3456	4152
1247_vm	16886_fsm	174	2618	2753_fsm	3456_fsm	4153
1249	16895	175	2618_fsm	2753_vm	3456_vm	4154
1249_fsm	16896	2187	2618_vm	2754	3466	4155
1249_vm	16914	2224	2619	2755	3467	4156
1255	16915	2228	2619_fsm	2756	3468	4157
1255_fsm	16915_fsm	2328	2619_vm	2757	3501	4158
1255_vm	16915_vm	2336	2620	2830	3502	4159
1259	16924	2337	2620_fsm	2831	3503	4160
1259_fsm	16925	2579	2650	2832	3534	4163
1259_vm	16938	2600	2651	2833	3541	4164
13476	16939	2600_fsm	2652	2834	3541_fsm	4165
13476_fsm	16939_fsm	2600_vm	2653	2835	3541_vm	4166
13476_vm	16939_vm	2601	2654	2836	3542	4167
13479	16949	2601_fsm	2655	2836_fsm	3574	4168
13480	16950	2601_vm	2656	2836_vm	3575	4169
13481	16950_fsm	2602	2657	2837	3576	4170
13481_fsm	16950_vm	2602_fsm	2658	2838	3596	4171
13481_vm	16966	2602_vm	2659	2838_fsm	3597	4172
13484	16967	2603	2660	2838_vm	3598	4173
13485	16981	2603_fsm	2661	2839	3599	4174
13486	16982	2603_vm	2662	2840	3600	5002
13486_fsm	16983	2604	2663	2840_fsm	3600_fsm	548
13486_vm	16984	2604_fsm	2664	2840_vm	3600_vm	549
13489	17003	2605	2665	2841	3601	6102
13490	17005	2605_fsm	2666	2995	3601_fsm	6104
13491	17007	2605_vm	2667	2996	3601_vm	6106
13491_fsm	17009	2606	2668	3079	3602	6110
13491_vm	17011	2606_fsm	2669	3079_fsm	3602_fsm	6111
13494	17013	2606_vm	2670	3079_vm	3602_vm	6112
13495	17015	2607	2673	3080	3603	6113
1417	17017	2607_fsm	2674	3081	3603_fsm	6116

1418	17019	2607_vm	2675	3085	3603_vm	6117
16798	17021	2608	2678	3118	3604	6175
16799	17023	2608_fsm	2679	3119	3605	6176
16799_fsm	17025	2608_vm	2680	3164	3606	6228
16815	17027	2609	2681	3256	3607	6229
16816	17029	2609_fsm	2682	3257	3608	6237
16816_fsm	17031	2609_vm	2683	3258	3609	6238
16825	17033	2610	2684	3350	3712	6239
16826	17034	2610_fsm	2685	3351	3764	6351
16834	17035	2610_vm	2686	3379	3764_fsm	6352
16835	17036	2611	2687	3380	3764_vm	826
16835_fsm	17037	2612	2688	3381	3766	827
16852	17038	2612_fsm	2689	3394	3767	828
16853	17039	2612_vm	2690	3394_fsm	3997	pg_filenode.map
16854	17040	2613	2691	3394_vm	4143	pg_internal.init
16854_fsm	17041	2615	2692	3395	4144	PG_VERSION
16854_vm	17042	2615_fsm	2693	3429	4145	
16861	17043	2615_vm	2696	3430	4146	
16861_fsm	17044	2616	2699	3431	4147	
16873	17045	2616_fsm	2701	3433	4148	

11. Affichage de PGDATA pg18_a

```
[11]: !sudo podman exec pg18_a ls -lrt /var/lib/postgresql/18/docker
```

```
total 128
drwx----- 2 postgres postgres 4096 Nov 17 09:49 pg_commit_ts
-rw----- 1 postgres postgres   3 Nov 17 09:49 PG_VERSION
drwx----- 2 postgres postgres 4096 Nov 17 09:49 pg_twophase
drwx----- 2 postgres postgres 4096 Nov 17 09:49 pg_stat_tmp
drwx----- 2 postgres postgres 4096 Nov 17 09:49 pg_snapshots
drwx----- 2 postgres postgres 4096 Nov 17 09:49 pg_serial
drwx----- 2 postgres postgres 4096 Nov 17 09:49 pg_replslot
drwx----- 2 postgres postgres 4096 Nov 17 09:49 pg_notify
drwx----- 4 postgres postgres 4096 Nov 17 09:49 pg_multixact
drwx----- 2 postgres postgres 4096 Nov 17 09:49 pg_dynshmem
-rw----- 1 postgres postgres   88 Nov 17 09:49 postgresql.auto.conf
-rw----- 1 postgres postgres 2681 Nov 17 09:49 pg_ident.conf
drwx----- 2 postgres postgres 4096 Nov 17 09:49 pg_xact
drwx----- 2 postgres postgres 4096 Nov 17 09:49 pg_subtrans
-rw----- 1 postgres postgres 32318 Nov 17 09:52 postgresql.conf
-rw----- 1 postgres postgres 5823 Nov 17 09:55 pg_hba.conf
drwx----- 4 postgres postgres 4096 Nov 17 10:41 pg_wal
-rw----- 1 postgres postgres   36 Nov 17 13:05 postmaster.opts
drwx----- 2 postgres postgres 4096 Nov 17 13:05 pg_stat
-rw----- 1 postgres postgres   99 Nov 17 13:05 postmaster.pid
drwx----- 5 postgres postgres 4096 Nov 17 13:57 base
drwx----- 2 postgres postgres 4096 Nov 17 13:58 pg_tblspc
```

```
drwx----- 2 postgres postgres 4096 Nov 17 14:02 global
drwx----- 4 postgres postgres 4096 Nov 17 14:22 pg_logical
```

```
[17]: !sudo podman ps -a
```

```
CONTAINER ID   IMAGE                                     COMMAND
CREATED        STATUS          PORTS
NAMES
74d35867d4a5   mcr.microsoft.com/mssql/server:2019-latest
/opt/mssql/bin/sq... 2 days ago      Exited (137) 2 hours ago
0.0.0.0:1435->1433/tcp      sql2019
f48bfcdbf0d6   docker.io/docker4chan/oracle-19c-prebuilt:19.3.0-ee /bin/sh -c
exec $... 2 days ago      Exited (137) 2 hours ago 0.0.0.0:2222->22/tcp,
0.0.0.0:1524->1521/tcp      oracle19c
8a751af75162   docker.io/library/postgres:18                postgres
4 hours ago    Up 31 minutes ago      0.0.0.0:5433->5432/tcp
pg18_a
a768998d6b02   docker.io/library/postgres:18                postgres
4 hours ago    Exited (0) 2 hours ago 0.0.0.0:5434->5432/tcp
pg18_b
3fe55f8c3083   docker.io/library/postgres:18                postgres
4 hours ago    Exited (0) 2 hours ago 0.0.0.0:5435->5432/tcp
pg18_c
```

```
[ ]: %config SqlMagic.displaylimit = None
```

```
[ ]: %%sql
select * from actor
```

```
[9]: !sudo podman ps -a
```

```
CONTAINER ID   IMAGE                                     COMMAND
CREATED        STATUS          PORTS
NAMES
74d35867d4a5   mcr.microsoft.com/mssql/server:2019-latest
/opt/mssql/bin/sq... 17 hours ago      Exited (0) 16 hours ago
0.0.0.0:1435->1433/tcp      sql2019
f48bfcdbf0d6   docker.io/docker4chan/oracle-19c-prebuilt:19.3.0-ee /bin/sh -c
exec $... 17 hours ago      Exited (137) 16 hours ago 0.0.0.0:2222->22/tcp,
0.0.0.0:1524->1521/tcp      oracle19c
6d4bf284356d   docker.io/library/postgres:18                postgres
14 minutes ago  Up 14 minutes ago      0.0.0.0:5433->5432/tcp
pg18_a
```

```
[10]: !sudo podman exec -it pg18_a psql -U postgres -d dvdrental -c "\dt"
```

```

List of tables
Schema | Name      | Type | Owner

```

```

-----+-----+-----+-----
public | actor          | table | postgres
public | address         | table | postgres
public | category        | table | postgres
public | city            | table | postgres
public | country         | table | postgres
public | customer        | table | postgres
public | film            | table | postgres
public | film_actor      | table | postgres
public | film_category   | table | postgres
public | inventory       | table | postgres
public | language        | table | postgres
public | payment         | table | postgres
public | rental          | table | postgres
public | staff           | table | postgres
public | store           | table | postgres
(15 rows)

```

```

[20]: query = """
SELECT datname, pg_tablespace.spcname AS tablespace
FROM pg_database
JOIN pg_tablespace ON pg_database.dattablespace = pg_tablespace.oid
WHERE datname = 'dvdrental';
"""

!sudo podman exec -it pg18_a psql -U postgres -d dvdrental -c "{query}"

```

```

    datname | tablespace
-----+-----
dvdrental | tbs_dvdrental
(1 row)

```

```

[18]: # Configuration : nom du conteneur et de la table
container_name = "pg18_a"    # change le nom si besoin
table_name = "actor"         # change le nom de la table que tu veux explorer
limit = 10                   # nombre de lignes à afficher

# Lister tous les conteneurs Podman
print("=== Conteneurs Podman ===")
!sudo podman ps -a

# Lister toutes les tables dans la base dvdrental
print(f"\n=== Tables dans dvdrental (conteneur {container_name}) ===")
!sudo podman exec -i {container_name} psql -U postgres -d dvdrental -c "\dt"

# Afficher les premières lignes de la table choisie

```



```
print(f"\n=== {limit} premières lignes de {table_name} ===")
!sudo podman exec -i {container_name} psql -U postgres -d dvdrental -c "SELECT_
↳* FROM {table_name} LIMIT {limit};"
```

=== Conteneurs Podman ===

CONTAINER ID	IMAGE	COMMAND
74d35867d4a5	mcr.microsoft.com/mssql/server:2019-latest	
/opt/mssql/bin/sql2019	2 days ago	Exited (137) 3 hours ago
0.0.0.0:1435->1433/tcp		
f48bfcdbf0d6	docker.io/docker4chan/oracle-19c-prebuilt:19.3.0-ee	/bin/sh -c
exec \$...	2 days ago	Exited (137) 3 hours ago
0.0.0.0:1524->1521/tcp	oracle19c	
8a751af75162	docker.io/library/postgres:18	postgres
5 hours ago	Up 2 hours ago	0.0.0.0:5433->5432/tcp
pg18_a		
a768998d6b02	docker.io/library/postgres:18	postgres
5 hours ago	Up 15 minutes ago	0.0.0.0:5434->5432/tcp
pg18_b		
3fe55f8c3083	docker.io/library/postgres:18	postgres
5 hours ago	Up 14 minutes ago	0.0.0.0:5435->5432/tcp
pg18_c		

=== Tables dans dvdrental (conteneur pg18_a) ===

List of tables

Schema	Name	Type	Owner
public	actor	table	postgres
public	address	table	postgres
public	category	table	postgres
public	city	table	postgres
public	country	table	postgres
public	customer	table	postgres
public	film	table	postgres
public	film_actor	table	postgres
public	film_category	table	postgres
public	inventory	table	postgres
public	language	table	postgres
public	payment	table	postgres
public	rental	table	postgres
public	staff	table	postgres
public	store	table	postgres

(15 rows)

=== 10 premières lignes de actor ===

actor_id	first_name	last_name	last_update
1	Penelope	Guinness	2013-05-26 14:47:57.62
2	Nick	Wahlberg	2013-05-26 14:47:57.62
3	Ed	Chase	2013-05-26 14:47:57.62
4	Jennifer	Davis	2013-05-26 14:47:57.62
5	Johnny	Lollobrigida	2013-05-26 14:47:57.62
6	Bette	Nicholson	2013-05-26 14:47:57.62
7	Grace	Mostel	2013-05-26 14:47:57.62
8	Matthew	Johansson	2013-05-26 14:47:57.62
9	Joe	Swank	2013-05-26 14:47:57.62
10	Christian	Gable	2013-05-26 14:47:57.62

(10 rows)

```
[ ]: !sudo podman exec -it pg18_a psql -U postgres -d dvdrental -c "\1"
```

List of databases						
Name	Owner	Encoding	Locale Provider	Collate	Ctype	
Locale	ICU Rules	Access privileges				
dvdrental	postgres	UTF8	libc	en_US.utf8	en_US.utf8	
postgres	postgres	UTF8	libc	en_US.utf8	en_US.utf8	
template0	postgres	UTF8	libc	en_US.utf8	en_US.utf8	
	=c/postgres					
	postgres=CTc/postgres					
template1	postgres	UTF8	libc	en_US.utf8	en_US.utf8	
	=c/postgres					
	postgres=CTc/postgres					

(4 rows)

(END)

12. Préparer un répertoire pour le tablespace pour dvdrental

```
sudo podman exec -u root pg18_a mkdir -p /var/lib/postgresql/18/tbs_dvdrental
sudo podman exec -u root pg18_a chown postgres:postgres /var/lib/postgresql/18/tbs_dvdrental
```

13. Créer une base dvdrental utilisant ce tablespace

```
sudo podman exec -it pg18_a psql -U postgres

CREATE TABLESPACE tbs_dvdrental
LOCATION '/var/lib/postgresql/18/tbs_dvdrental';
```

```
CREATE DATABASE dvdrental
WITH TABLESPACE = tbs_dvdrental;
```

```
[ ]: %config SqlMagic.displaylimit = 10
```

```
[ ]: !sudo podman exec -it pg18_a psql -U postgres -d dvdrental -c "SELECT c.relname,
↳AS index_name, c.reltablespace FROM pg_index i JOIN pg_class c ON i.
↳indexrelid = c.oid;"
```

index_name	reltablespace
pg_toast_16835_index	0
pg_toast_16967_index	0
actor_pkey	0
address_pkey	0
category_pkey	0
city_pkey	0
country_pkey	0
customer_pkey	0
film_actor_pkey	0
film_category_pkey	0
film_pkey	0
inventory_pkey	0
language_pkey	0
payment_pkey	0
rental_pkey	0
staff_pkey	0
store_pkey	0
film_fulltext_idx	0
idx_actor_last_name	0
idx_fk_address_id	0
idx_fk_city_id	0

```
:
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```